	Sanjivani Rural Educational Society's **SANJIVANI COLLEGE OF ENGINEERING** **(An Autonomous Institution)** **Koregaon – 423 603, Maharashtra.**	**ACAD-F-15H**
Academic Year: 2026- 2027		**IoT**
SEM-V	**Department: Computer Engineering**	**Date of Preparation: 21 / 07 /2026**

Group ID: 03

Activity Title: AI-Based Non-Communicable Disease Prediction and Smart Medicine Reminder System using IoT

Introduction:

Chronic (non-communicable) diseases such as Diabetes, Hypertension, Heart Disease, and kidney disease require patients to take medicines regularly and without fail. In practice, patients frequently forget their doses, family members remain unaware of whether medicines were taken, doctors have no reliable way to track medicine adherence, and medical reports are often too technical for patients to interpret on their own.

At the district level, Community Health Officers (CHOs) also lack a system to monitor patients across an entire region. The "AI-Based Non-Communicable Disease Prediction and Smart Medicine Reminder System" is designed to address all of these problems together by integrating IoT-based automatic medicine dispensing, Optical Character Recognition (OCR) for report digitisation, Artificial Intelligence (AI) for disease-risk prediction, and a Web Dashboard for role-based monitoring by patients, doctors, family members/nurses, and CHOs.

Existing System:

Presently, medicine adherence for chronic disease patients is managed through:

- Manual reminders set by patients or family members (alarms, verbal reminders)
- Ordinary pill boxes with no sensing, alerting, or logging capability
- Manual, paper-based recording of medicine intake, if recorded at all
- Physical submission of lab/medical reports that are read and interpreted manually by doctors
- No centralized platform through which doctors, family members, or health officers can track a patient's status remotely

Limitations of the Existing Approach:

- No automatic reminder or dispensing mechanism at the exact prescribed time
- No way to confirm whether a dispensed dose was actually taken
- Family members are not automatically informed when a dose is missed
- No digital medicine history, making adherence patterns hard to analyse
- Medical reports must be read manually, which is time-consuming and error-prone
- No predictive assessment of disease risk from report values
- No mechanism for CHOs to monitor medicine adherence and disease statistics across many patients or villages
- Doctors get delayed, incomplete visibility into patient recovery and behaviors

Proposed System:

- The proposed system uses IoT, AI, OCR, and a web dashboard to automate medicine reminders and disease monitoring for chronic disease patients. It demonstrates:
- IoT-based automatic medicine dispensing using ESP32, RTC, servo motors, buzzer, LEDs, and an IR sensor/push button
- Real-time schedule delivery and dose confirmation between the dispenser and cloud database
- Automatic reminders and SMS alerts to family members/nurses when a dose is missed
- OCR-based extraction of values from uploaded medical reports using Tesseract
- AI-based prediction of Diabetes, Heart Disease, Hypertension, Kidney Disease, and Stroke risk
- Role-based web dashboard for Patient, Doctor, Family Member/Nurse, and CHO
- Medicine adherence tracking and recovery monitoring through historical data comparison
- District-level, village-wise patient and disease monitoring for CHOs
- The system ensures reliable medicine adherence by combining sensor-based confirmation, cloud storage, and AI-driven analysis, while giving every stakeholder — from patient to district health officer — a real-time, role-specific view of health status.

Objectives:

- The main objectives of this project are:
- To remind patients to take their medicines on time
- To automatically dispense the correct medicine at the scheduled time
- To notify family members whenever a medicine dose is missed
- To store a complete, searchable medicine history for every patient
- To allow patients to upload medical reports for digital analysis
- To analyse uploaded reports using OCR and AI
- To predict the risk of major non-communicable diseases from report data
- To help doctors monitor prescriptions, adherence, and patient recovery
- To help Community Health Officers monitor all patients at the district level
- To reduce medication errors and missed doses through automated hardware-level dispensing
- To provide a centralised, role-based platform connecting patients, doctors, family members, and health officers

Scope:

- The scope of this project includes:
- Role-based access for Patients, Doctors, Family Members/Nurses, and CHOs
- Automated, IoT-driven medicine reminding and dispensing for chronic diseases such as Diabetes, Hypertension, Heart Disease, and kidney disease
- Digitisation and AI-based interpretation of medical reports through OCR and Machine Learning
- Adherence tracking, recovery monitoring, and nurse-alerting for individual patients
- District-level, village-wise disease and adherence monitoring for public health officers
- A design that can be extended to additional non-communicable diseases and larger patient populations in the future

Importance of Project:

- Non-communicable diseases are a major and growing public health concern, and poor medicine adherence significantly worsens patient outcomes. This project is important because it:
- Reduces missed doses and the health complications that follow from irregular medication
- Keeps family members and doctors promptly informed about a patient's medicine status
- Uses AI and OCR to make medical reports understandable and to flag disease risk early, encouraging timely doctor visits
- Gives doctors objective, data-backed visibility into adherence and recovery trends rather than relying on patient self-reporting
- Enables Community Health Officers to monitor disease burden and adherence across an entire district, supporting public health planning
- Combines IoT, AI, OCR, and web-dashboard technologies into a single practical system, building strong, industry-relevant skills for the team in embedded systems, machine learning, and full-stack development

Expected Input:

- The system accepts inputs from IoT hardware sensors as well as from users through the web dashboard.
- Hardware/Sensor Inputs:
- Real-time clock (RTC) readings, continuously checked against the stored schedule to trigger dispensing at the exact prescribed time
- Push button press or IR sensor detection, confirming that the patient has physically picked up the dispensed medicine
- Medicine schedule data received by the ESP32 over Wi-Fi, including medicine name, slot number, and dosage timing
- Device status signals (Wi-Fi connectivity, servo/buzzer operation) used to confirm the dispenser is functioning correctly
- User-Provided Inputs:
- Medicine schedule entered by the doctor, specifying medicine name, dosage, frequency, and timing (e.g., Metformin, Morning, 8:00 AM)
- Patient-uploaded medical reports (Blood Test, CBC, Sugar Report, etc) submitted as PDF or image files for OCR and AI analysis
- Login credentials and role information for Patient, Doctor, Family Member/Nurse, and CHO, used for role-based access to the dashboard
- Nurse-assignment details entered by the doctor, linking a specific nurse to a specific patient for

missed-dose alerts

- Patient and family contact details, used to route SMS/app notifications correctly
- These inputs together allow the system to trigger accurate, time-based dispensing, confirm medicine intake, digitise medical reports, and route the right information and alerts to the right user role.

Expected Output:

- Automatic opening of the correct pill slot with buzzer and LED alert at the scheduled time
- Dashboard status updates showing "Taken" or "Missed" with patient ID, medicine name, scheduled time, taken time, and delay
- Automatic reminder to the patient and SMS alert to the family member/nurse when a dose is missed
- Complete, date-wise medicine history for every patient
- OCR-extracted, structured values (e.g., Blood Sugar, BP, HbA1c) from uploaded reports
- AI-predicted risk percentages for Diabetes, Heart Disease, Hypertension, Kidney Disease, and Stroke
- Medicine adherence percentage for each patient
- Recovery-trend graphs comparing successive report values over time
- District-level CHO dashboard showing total patients, high-risk patients, overall adherence percentage, most common disease, and village-wise statistics

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